SPHERE PACKINGS WHOSE TANGENCY POLYTOPES TILE A PRESCRIBED OPEN SUBSET OF $\mathbb{R}^3$

ABSTRACT. For every nonempty open set $\Omega \subseteq \mathbb{R}^3$ and every $\varepsilon > 0$, we construct a countably infinite packing by positive-radius closed balls whose common-tangent-halfspace polytopes form a locally finite, face-to-face tiling of exactly Ω. Ball tangencies are precisely the facet adjacencies of the tiling. Each polytope has at most twelve facets and lies between its associated ball and the concentric ball of $\sqrt{10}$ times the radius; all radii are at most ε. The construction uses three local modifications of the rhombic-dodecahedral lattice tiling, followed by stabilization away from the complement of Ω.

1. Statement and terminology

Write $B(x,r)$ and $\overline{B}(x,r)$ for the open and closed Euclidean balls. A polytope will always mean a bounded, closed, convex polytope with nonempty interior. A *locally finite, face-to-face tiling* of an open set $D \subseteq \mathbb{R}^3$ is a family $\{Q_v\}$ of polytopes such that

$$Q_v \subseteq D, \qquad \bigcup_v Q_v = D, \qquad \operatorname{int} Q_v \cap \operatorname{int} Q_w = \varnothing \quad (v \neq w), \tag{1.1}$$

every compact subset of D meets finitely many tiles, and every nonempty intersection of distinct tiles is a face of both. A facet means a two-dimensional face. Given a point $v \in \operatorname{int} Q_v$ for each tile, the associated tiling graph joins v and w when $Q_v \cap Q_w$ is a facet. It is *orthogonal* if $w - v$ is perpendicular to $\operatorname{aff}(Q_v \cap Q_w)$ for every adjacent pair. This is the orthogonality condition of [BG25, Section 1.1].

A *sphere packing* is a family $\{\overline{B}(v, r_v) : v \in V\}$, with $r_v > 0$, whose ball interiors are pairwise disjoint. Put $v \sim w$ when $|v - w| = r_v + r_w$. For such a pair, define

$$H^-_{v,w} = \left\{ x \in \mathbb{R}^3 : (x - v) \cdot \frac{w - v}{|w - v|} \leq r_v \right\}. \tag{1.2}$$

Its boundary is the common tangent plane, and it contains the ball at v. The *tangency polytope*, when the intersection is a polytope, is

$$P_v = \bigcap_{w \sim v} H^-_{v,w}. \tag{1.3}$$

Theorem 1.1. *Let $\Omega \subseteq \mathbb{R}^3$ be nonempty and open, and let $\varepsilon > 0$. There is a countably infinite sphere packing $\{\overline{B}(v, r_v) : v \in V\}$ such that the polytopes (1.3) form a locally finite, face-to-face tiling of exactly Ω. For distinct v, w, the balls are tangent if and only if $P_v \cap P_w$ is a facet, and that facet lies in their common tangent plane. Moreover, every P_v has at most twelve facets and*

$$0 < r_v \leq \varepsilon, \qquad \overline{B}(v, r_v) \subseteq P_v \subseteq \overline{B}(v, \sqrt{10}\, r_v). \tag{1.4}$$

In particular, the tangency graph is the orthogonal tiling graph.

Coverage in this statement is by the polytopes, not by the balls. Local finiteness is relative to Ω; accumulation at its boundary is allowed. No regularity or connectedness assumption on Ω is imposed.

2. Tangential tilings and power cells

A polytope is *circumscribed* about a ball if it contains the ball and every facet plane is tangent to the ball. Two such balls have *matching contacts* across a shared facet if their contact points with its plane coincide. We call a locally finite, face-to-face tiling supplied with circumscribed positive-radius balls and matching contacts a *paired tiling*. This definition applies to any open set.

Lemma 2.1. *In a paired tiling of an open set D, the balls form a sphere packing. Their tangencies are exactly the facet adjacencies, and each tile is its ball's tangency polytope.*

Proof. The interior of a ball contained in a polytope lies in the interior of that polytope, so distinct ball interiors are disjoint. A ball cannot touch a face of codimension at least two of a containing polytope: two distinct facet planes would then support the ball at the same point, contrary to uniqueness of its supporting plane there.

Suppose two balls are tangent at t. The point t is on both tile boundaries. Otherwise interior points of one ball arbitrarily close to t would belong to the interior of the other tile. The preceding observation places t in the relative interior of a facet of each tile. The face-to-face condition identifies these facets. Conversely, matching contacts across a facet give ball tangency.

Every facet F of a tile Q has a neighboring tile. Indeed, for $t \in \operatorname{relint} F$, choose points outside Q converging to t in a compact neighborhood contained in D. Coverage and local finiteness give a subsequence contained in one other tile Q'. Closedness gives $t \in Q'$, and the face-to-face condition then yields $Q \cap Q' = F$. Thus the halfspaces in (1.3) are exactly the facet halfspaces of Q. □

For a ball with center a and radius r, write

$$\pi_{a,r}(x) = |x-a|^2 - r^2, \qquad \ell_{a,r}(x) = 2a \cdot x - |a|^2 + r^2. \tag{2.1}$$

Its power cell in a family of balls is the set where its power is no greater than that of every other ball.

Lemma 2.2. *A paired tiling $\{(\overline{B}(a,r_a), Q_a)\}$ of all of $\mathbb{R}^3$ is the power diagram of its balls:*

$$Q_a = \{x : \pi_{a,r_a}(x) \le \pi_{b,r_b}(x) \text{ for every } b\}. \tag{2.2}$$

In particular the minimum power is attained at every point.

Proof. Let Q_a and Q_b share a facet, and let n be its unit normal pointing toward Q_b. Matching contacts give

$$b - a = (r_a + r_b)n, \qquad \ell_{b,r_b}(x) - \ell_{a,r_a}(x) = 2(r_a + r_b)(n \cdot (x-a) - r_a).$$

Thus these affine functions agree on the common facet plane.

The affine pieces also agree at all lower-dimensional intersections. To see this at a point p, choose a sufficiently small ball around p so that every tile meeting the ball contains p; this is possible by local finiteness and closedness. The incident tiles can be joined through facets containing p, by a generic segment between interior points in that ball. Agreement propagates along this chain. Consequently the prescription $f = \ell_{a,r_a}$ on Q_a defines a continuous, locally piecewise-affine function on $\mathbb{R}^3$.

On a line segment crossing only facet interiors, the slope jump on crossing from Q_a to Q_b is $2(r_a + r_b)n \cdot v > 0$, where v is the direction of traversal. Such segments are dense among all segments, since a compact neighborhood meets only finitely many tile edges and vertices. Continuity gives convexity on every segment. Hence f is convex on all of $\mathbb{R}^3$.

Each affine piece is a global supporting affine function of f, so $f(x) = \max_a \ell_{a,r_a}(x)$. Its equality set with ℓ_{a,r_a} is exactly Q_a. Indeed, if equality also held at $x \notin Q_a$, convexity would give equality on the convex hull of x and a small ball in $\operatorname{int} Q_a$. That hull contains a nonempty open set outside the closed set Q_a. Local finiteness then gives an open subset of another tile on which two distinct affine

pieces coincide, an impossibility since their gradients are twice their distinct centers. Equation (2.2) follows. □

We use Lemma 2.2 only for whole-space tilings, before taking a limit on the prescribed open set. The following criterion includes the comparisons needed for simultaneous insertions.

Lemma 2.3 (Insertion criterion)**.** *Let $L \subset \mathbb{R}^3$ be a lattice, let $r > 0$, and let $\mathcal{R} \subset L$ be a finite set of vectors of length $2r$. Start from a paired tiling of $\mathbb{R}^3$. Let $A \subset L$ be a set of new centers, each assigned radius r. Suppose the enlarged family of balls has disjoint interiors. For each $c \in A$, let*

$$W_c = \bigcap_{u \in N_c} \{x : u \cdot (x - c) \le 2r^2\}, \qquad N_c \subseteq \mathcal{R}, \tag{2.3}$$

be a polytope, with $W_c \subseteq \overline{B}(c, M)$ for a constant M independent of c. Assume the following:

1. *For every $u \in N_c$, the center $c + u$ belongs either to A or to an old ball of radius r.*
2. *On W_c, $\pi_{c,r}$ is no greater than the old minimum power and is no greater than $\pi_{b,r}$ for every $b \in A$.*

Then adding all the new balls and taking power cells gives another paired tiling. Each new cell is exactly W_c, and every old cell only shrinks. If all W_c lie in a union U of a specified subfamily of old tiles, every old tile not in that subfamily remains unchanged.

Proof. The second hypothesis puts W_c in the new power cell of c. Outside W_c, one of the inequalities in (2.3) fails, and the neighbor supplied by the first hypothesis has strictly smaller power. Hence the new cell is exactly W_c.

The minimum among the old powers is attained by Lemma 2.2. For each fixed x, the new powers tend to infinity as their lattice centers tend to infinity. Thus the minimum over the enlarged family is also attained, and its power cells cover $\mathbb{R}^3$.

Old cells are intersected with new power halfspaces. They retain their balls: for any nonoverlapping balls, every point of one ball has nonpositive power for that ball and nonnegative power for the other. On a fixed old polytope, all sufficiently distant new sites have uniformly larger power than its own site. Only finitely many new inequalities can therefore affect that polytope. Hence old cells remain full-dimensional bounded polytopes. New cells contain their balls by (2.3), and local finiteness follows from old local finiteness and $W_c \subseteq \overline{B}(c, M)$. The power inequalities imply the face-to-face property: the intersection of two cells lies in their common supporting power-bisector plane and is a face of both.

If a facet of W_c is defined by $u \in N_c$, its points have minimum power also for the site $c + u$. The cell at $c + u$ therefore contains that facet and is its neighboring cell. The two equal-radius balls are tangent at $c + u/2$. An old–old facet is contained in an old shared facet, so has the same common tangent plane. The old contact point survives because both old balls remain in their cells. Every facet thus has matching ball contacts, proving the paired-tiling assertion.

Finally, at a point removed from an old cell, a minimizing new site has strictly smaller power than the old minimum. Such a point belongs to the union of the W_c. If an old tile not selected for U lost a boundary point, strict inequality and continuity would make it lose interior points as well. Its interior is disjoint from U, so this is impossible. Such a tile remains unchanged as a closed polytope. □

3. The lattice and two elementary polytopes

Set

$$L = \{z \in \mathbb{Z}^3 : z_1 + z_2 + z_3 \text{ is even}\}, \qquad \mathcal{R} = \{\pm e_i \pm e_j : 1 \le i < j \le 3\}, \qquad r_* = \frac{1}{\sqrt{2}}, \tag{3.1}$$

where the signs are independent. Define

$$P = \{x : u \cdot x \le 1 \text{ for all } u \in \mathcal{R}\} = \{x : |x_i| + |x_j| \le 1 \text{ for } i < j\}. \tag{3.2}$$

Lemma 3.1. *The polytope P is the Voronoi cell at the origin of L. Its vertices are*

$$\pm e_i \ (1 \le i \le 3), \qquad (\pm\tfrac{1}{2}, \pm\tfrac{1}{2}, \pm\tfrac{1}{2}), \tag{3.3}$$

and $\overline{B}(0, r_*) \subseteq P \subseteq \overline{B}(0, 1)$. The pairs

$$(\overline{B}(z, hr_*),\ z + hP), \qquad z \in hL, \tag{3.4}$$

form a paired tiling of $\mathbb{R}^3$ for every $h > 0$.

Proof. The vertex list follows from the inequalities in (3.2). At a vertex $x = \pm e_i$, the lattice Voronoi inequality $2x \cdot z \le |z|^2$ follows from $2|z_i| \le |z|^2$ for $z \in L$. For $|z_i| \ge 2$ this follows from $2|z_i| \le z_i^2$; when $|z_i| = 1$, parity forces at least one other nonzero coordinate. At any of the remaining vertices,

$$2x \cdot z \le \sum_i |z_i| \le \sum_i z_i^2.$$

Thus every lattice Voronoi inequality holds on P. Conversely, the inequalities corresponding to $\mathcal{R}$ are among those inequalities and define P. This proves the Voronoi assertion. The inclusions follow from the facet distances and (3.3). Across a facet indexed by $u \in \mathcal{R}$, the balls at 0 and u meet at $u/2$. Translation and scaling prove the last assertion. □

For $i \in \{1, 2, 3\}$, let

$$U_i = \{u \in \mathcal{R} : u_i = 0\}, \qquad K_i = \{y : |y_i| + |y_j| \le 1 \text{ for } j \ne i\}. \tag{3.5}$$

Lemma 3.2. *The vertices of K_i are $\pm e_i$ and the four vectors $\pm e_j \pm e_k$, where $\{i, j, k\} = \{1, 2, 3\}$. Moreover,*

$$K_i = \{y : 2y \cdot v \le |v|^2 \text{ for every } v \in L \setminus U_i\}. \tag{3.6}$$

It is circumscribed about $\overline{B}(0, r_*)$ and lies in $\overline{B}(0, \sqrt{2})$.

Proof. The vertex description follows by slicing at a fixed value of y_i: the remaining coordinates range independently over $[-(1 - |y_i|), 1 - |y_i|]$. At $y = \pm e_i$, the inequality in (3.6) follows as in Lemma 3.1. At a vertex $y \in U_i$, every $v \in L \setminus U_i$ satisfies $|v - y|^2 \ge 2$, since $v - y$ is a nonzero vector of L. Expanding this inequality gives $2y \cdot v \le |v|^2$. Conversely the constraints indexed by $\mathcal{R} \setminus U_i$ define K_i. Its facet planes have distance r_* from the origin, and its vertices have norm at most $\sqrt{2}$. □

4. A local refinement

Call a pair in (3.4) a *standard scale-h pair*.

Proposition 4.1 (Local refinement)**.** *Suppose a paired tiling of $\mathbb{R}^3$ contains the standard scale-h pair at every point of a set $S \subseteq hL$. These pairs can be replaced, leaving all other old pairs unchanged, by a paired tiling with the following properties:*

1. *Every new ball has radius $hr_*/3$. Every new polytope lies in $\bigcup_{z \in S}(z + hP)$.*
2. *Every new polytope has at most twelve facets and is contained in the concentric ball of $\sqrt{10}$ times its ball radius.*
3. *If $c \in (h/3)L$ and every $z \in hL$ with $|z - c| \le 10h$ belongs to S, the new tiling contains the complete standard scale-$h/3$ pair at c.*

Proof. By scaling, it suffices to take $h = 3$. Thus $S \subseteq 3L$, and all new radii are r_*. We give three successive operations.

First operation: subdivision of selected cells. For a selected cell $3P$ centered at 0, use the central pair $(\overline{B}(0,r_*),P)$ and the twelve balls $\overline{B}(u,r_*)$, $u\in\mathcal{R}$. The polytope assigned to u is

$$C_u=\{x\in 3P : u\cdot x\ge 1,\quad u\cdot x\ge v\cdot x \text{ for all } v\in\mathcal{R}\}. \tag{4.1}$$

For $u=e_1+e_2$, this is

$$C_u=\{x : 1\le x_1+x_2\le 3,\quad x_1\ge |x_3|,\quad x_2\ge |x_3|\}. \tag{4.2}$$

The other cases follow by signed coordinate permutations. These cells partition $3P$: outside P, choose a root maximizing $u\cdot x$. Equivalently, they are the equal-radius Voronoi cells of $\{0\}\cup\mathcal{R}$, clipped to $3P$, so meet face-to-face.

Each of the six facet planes in (4.2) has distance r_* from $u=(1,1,0)$. Across its inner facet the neighbor is the central ball. Across, for example, $x_1=x_3$, the neighbor center is $(0,1,1)$; the difference is a root. Signed coordinate permutations give the other internal contacts.

On the outer facet $u\cdot x=3$, the new ball touches at $3u/2$, exactly the old radius-$3r_*$ ball's contact point. This outer facet is unchanged as a whole. Its intersections with other new cells are its old lower-dimensional faces. Thus the induced subdivision of the old cell boundary is unchanged, and the replacements match every unselected neighboring tile as well as one another. This proves that the first operation gives a paired tiling.

The vertex set of (4.2) consists of the four points

$$(1,0,0),\ (0,1,0),\ (\tfrac12,\tfrac12,\tfrac12),\ (\tfrac12,\tfrac12,-\tfrac12)$$

and their triples. Their distances from $(1,1,0)$ are at most $\sqrt{5}=\sqrt{10}\,r_*$. The central cell satisfies a stronger bound by Lemma 3.1.

After translating these replacements to all $z\in S$, every cell in the selected region has a center in L and radius r_*. Outside that region all old pairs are unchanged.

Second operation: insertion at six-cell vertices. Let

$$\mathcal{Q}=\{3m : m\in\mathbb{Z}^3,\ m_1+m_2+m_3 \text{ is odd}\}.$$

Call $q\in\mathcal{Q}$ eligible when all six points $q\pm 3e_i$, $1\le i\le 3$, belong to S. At each eligible q, insert the six balls with centers

$$c=q+\sigma e_i,\qquad 1\le i\le 3,\quad \sigma\in\{-1,1\}, \tag{4.3}$$

and assign to such a center the proposed cell

$$W_c=c+K_i. \tag{4.4}$$

We verify Lemma 2.3, including its simultaneous comparisons.

The cell W_c is contained in the selected parent cell with center

$$z=q+3\sigma e_i. \tag{4.5}$$

Indeed, relative to z, its vertices are $-3\sigma e_i$, $-\sigma e_i$, and $-2\sigma e_i\pm e_j\pm e_k$, all of which satisfy the inequalities of $3P$.

Every ball introduced so far inside the selected region has radius r_* and center in L. The new centers in (4.3) also lie in L by parity. They are distinct: their single nonzero residue determines i, σ, and then q. They were not produced by the first operation, whose centers have zero or two nonzero residues modulo 3. Since the shortest nonzero vector of L has length $2r_*$, these balls have disjoint interiors. Their containment in selected parent cells gives disjointness from all unselected old ball interiors.

The current tiles covering the parent in (4.5) have centers $b=z$ or $b=z+v$, $v\in\mathcal{R}$. Hence $b-c$ has ith coordinate among $\sigma,2\sigma,3\sigma$, and in particular does not belong to U_i. Lemma 3.2 therefore gives $\pi_{c,r_*}\le\pi_{b,r_*}$ on W_c. At each point of W_c, choose a current tile in that parent containing the

point. Lemma 2.2 shows that its power is the current minimum. This proves comparison with every current ball.

Each center in (4.3) has exactly one coordinate nonzero modulo 3. If c' is another such center, then $c' - c \notin U_i$: adding a vector of U_i to c would produce three coordinates nonzero modulo 3. Thus Lemma 3.2 also proves comparison with every other simultaneous insertion on W_c.

Finally, the eight facet constraints of K_i correspond to roots with nonzero ith coordinate. Their neighbors are

$$c - \sigma e_i + \tau e_j = q + \tau e_j, \qquad c + \sigma e_i + \tau e_j = z - \sigma e_i + \tau e_j, \qquad j \neq i, \quad \tau \in \{-1, 1\}.$$

The first is another insertion at the same q; the second was produced by the first operation in the parent $z + 3P$. All the hypotheses of Lemma 2.3 hold. It produces exactly the cells (4.4), shrinks the previous selected cells, and leaves the unselected region unchanged. These new cells have circumradius at most $\sqrt{2}$ about their centers.

Third operation: insertion of four-point clusters. For $b \in L$, let $k(b)$ be the number of its coordinates nonzero modulo 3. If every coarse cell were selected, the first operation would produce exactly the sites with $k(b) = 0$ or 2, and the second would produce exactly those with $k(b) = 1$. For $k(b) = 2$, write uniquely $b = z + u$ with $u \in \mathcal{R}$ obtained by representing each nonzero residue by 1 or -1; parity gives $z \in 3L$. For $k(b) = 1$, write uniquely $b = q + \sigma e_i$ with $q \in \mathcal{Q}$. The case $k(b) = 0$ is immediate from parity.

The remaining sites form disjoint sets

$$T_m = (3m + \{1, 2\}^3) \cap L, \qquad q_m = 3m + (\tfrac{3}{2}, \tfrac{3}{2}, \tfrac{3}{2}), \qquad m \in \mathbb{Z}^3. \tag{4.6}$$

Each T_m has four points. Two distinct points in it differ in exactly two coordinates, so their difference is a root. If $c \in T_m$ and $u \in \mathcal{R}$, then $c + u$ either has $k(c + u) \leq 2$ or belongs to the same T_m. Indeed, a root changes two coordinates by one, and preserving nonzero residues forces each changed coordinate to switch between $3m_j + 1$ and $3m_j + 2$.

Call m eligible when

$$\{z \in 3L : |z - q_m| \leq 12\} \subseteq S. \tag{4.7}$$

For every eligible m, insert all four balls at T_m, with proposed cells

$$W_c = c + P, \qquad c \in T_m. \tag{4.8}$$

The residue descriptions show that these centers do not coincide with any earlier center in the selected region, and that different clusters are disjoint. Again we verify Lemma 2.3.

First, $|c - q_m| = \sqrt{3}/2$ and $P \subseteq \overline{B}(0, 1)$. Any coarse cell meeting $c + P$ has its center within distance $\sqrt{3}/2 + 1 + 3 < 5$ of q_m, so was selected. Hence $c + P$ is covered by current cells having centers in L and radius r_*. The lattice Voronoi property of P proves that π_{c,r_*} is no greater than the old minimum power on $c + P$. It also proves comparison with all the other new sites, which belong to L. Containment of the proposed balls in the selected region and lattice separation prove nonoverlap with all current and new balls.

It remains to supply every root-neighbor $b = c + u$. If $k(b) = 3$, it belongs to the same cluster. Otherwise

$$|b - q_m| \leq \sqrt{3}/2 + \sqrt{2}.$$

If $k(b) = 0$ or 2, its first-operation parent is within a further distance $\sqrt{2}$. If $k(b) = 1$, write $b = q' + \sigma e_i$; all six parents needed to make q' eligible are within a further distance $1 + 3$ of b. In either case every required parent is within distance 12 of q_m, so was selected. Thus all root-neighbors are present. Lemma 2.3 applies and gives exactly (4.8), without changing any unselected old pair.

Bounds and complete fine cells. All the new facet normals in the three operations are positive multiples of roots in $\mathcal{R}$. Later insertions only cut earlier new cells by planes with these normals. Since there are twelve possible oriented normals, every new polytope has at most twelve facets. The circumradius bound follows from the first-operation bound $\sqrt{5}$, the second-operation bound $\sqrt{2}$, and the third-operation bound 1, because taking power cells only shrinks previously produced cells. All modifications are contained in the selected coarse region. Local finiteness follows from Lemma 2.3 and the locally finite first subdivision.

For completeness, a site $b \in L$ is produced whenever all coarse centers within distance 13 of b are selected. The required distances are at most $\sqrt{2}$ in the first operation, at most 4 in the second, and at most $12 + \sqrt{3}/2 < 13$ in the third. Thus selecting every coarse center within distance 30 of c produces c and all twelve sites $c + u$, $u \in \mathcal{R}$, since $13 + \sqrt{2} < 30$.

The resulting power cell at c is contained in $c + P$ by these root-neighbor inequalities. Conversely every coarse cell meeting $c + P$ has its center within distance 4 of c and is selected. At each point of $c + P$, a current equal-radius lattice cell therefore attains the minimum power. The lattice Voronoi inequalities show that c has no larger power there. Its cell is exactly $c + P$. Rescaling from $h = 3$ proves all three assertions. □

5. Refinement near the complement

Let $F = \mathbb{R}^3 \setminus \Omega$. If $F = \varnothing$, the standard tiling at scale $\sqrt{2}\varepsilon$ proves Theorem 1.1. Assume henceforth that $F \neq \varnothing$, and put

$$h_n = \sqrt{2}\varepsilon\, 3^{-n}, \qquad L_n = h_n L. \tag{5.1}$$

We construct paired tilings $\mathcal{T}_n$ of all of $\mathbb{R}^3$ satisfying

$$z \in L_n, \quad \operatorname{dist}(z, F) \le 30h_n \quad \Longrightarrow \quad (\overline{B}(z, h_n r_*), z + h_n P) \in \mathcal{T}_n. \tag{5.2}$$

Start with the standard scale-h_0 tiling. Given $\mathcal{T}_n$, apply Proposition 4.1 to

$$S_n = \{z \in L_n : \operatorname{dist}(z, F) \le 20h_n\}. \tag{5.3}$$

These full pairs are available by (5.2). If $c \in L_{n+1}$ and $\operatorname{dist}(c, F) \le 30h_{n+1} = 10h_n$, every $z \in L_n$ with $|z - c| \le 10h_n$ satisfies $\operatorname{dist}(z, F) \le 20h_n$. The last part of Proposition 4.1 gives (5.2) at stage $n + 1$.

Every standard scale-h_n polytope lies within distance h_n of its center. Consequently,

(5.4)
Every old pair removed and every new pair introduced in $\mathcal{T}_n \longrightarrow \mathcal{T}_{n+1}$ has its entire polytope in

$$\{x \in \mathbb{R}^3 : \operatorname{dist}(x, F) \le 21h_n\}.$$

All other complete pairs remain unchanged. Each $\mathcal{T}_n$ is locally finite, its radii lie in $[h_n r_*, \varepsilon]$, and all its pairs satisfy the facet and shape bounds of Theorem 1.1.

Retain precisely the eventually unchanged pairs:

$$\mathcal{T}_\infty = \bigcup_{N \ge 0} \bigcap_{n \ge N} \mathcal{T}_n. \tag{5.5}$$

Lemma 5.1. *For each compact $K \subseteq \Omega$, the collections of pairs whose polytopes meet K are identical in $\mathcal{T}_\infty$ and in $\mathcal{T}_N$ for every sufficiently large N. They are finite and cover K.*

Proof. For nonempty K, let $\delta = \operatorname{dist}(K, F) > 0$ and choose N such that $21h_N < \delta$. By (5.4), no subsequent operation removes or introduces a pair whose polytope meets K. Finite-stage local finiteness and coverage give the conclusion. The empty case is immediate. □

Lemma 5.2. *No polytope of $\mathcal{T}_\infty$ meets F.*

Proof. We show that any pair of $\mathcal{T}_n$ whose polytope Q meets F is removed at the next step. Fix $x \in Q \cap F$. Every standard scale-h_n lattice cell meeting $B(x, h_n)$ has its center within distance $2h_n$ of x, so occurs in $\mathcal{T}_n$ by (5.2). These finitely many standard cells cover $B(x, h_n)$.

Since Q is full-dimensional and convex, $\operatorname{int} Q \cap B(x, h_n)$ is nonempty and open, even when x lies on its boundary. A finite union of standard-cell boundaries cannot cover this open set. The interior of Q therefore meets the interior of one of those standard cells. Disjointness of tile interiors identifies Q, and hence its complete ball–polytope pair, with that standard pair. Its center is within distance h_n of F, so it belongs to S_n and its complete pair is replaced. An eventually unchanged pair cannot meet F. □

Proof of Theorem 1.1. Any two pairs of $\mathcal{T}_\infty$ coexist at a sufficiently late finite stage. Their ball centers are therefore distinct, so we may index the retained pairs by their centers v. Their polytopes have disjoint interiors and meet face-to-face, and their balls have matching contacts across each shared facet. Each retained pair inherits its positive radius, circumscribed polytope, and the bounds (1.4) from a finite stage.

Lemmas 5.1 and 5.2 show that these polytopes form a locally finite tiling of exactly Ω. Applying Lemma 2.1 identifies them with (1.3) and proves the asserted equivalence between tangencies and facet adjacencies. Matching contacts give orthogonality.

Each finite-stage tiling is countable, by local finiteness and exhaustion of $\mathbb{R}^3$ by compact balls. Hence (5.5) is countable. It is infinite, since a finite union of bounded closed polytopes is compact, whereas a nonempty open subset of $\mathbb{R}^3$ is not compact. □

Remark 5.3 (Tangency defect). If a polytope Q contains $\overline{B}(v, r)$ and h_G is the distance from v to the plane of a facet G, decomposition into pyramids gives

$$3|Q| - r\mathcal{H}^2(\partial Q) = \sum_{G \text{ facet of } Q} \mathcal{H}^2(G)(h_G - r) \geq 0.$$

The defect vanishes precisely when Q is circumscribed about the ball. Every intermediate pair in the construction has zero defect; the proof does not rely on showing that a positive-defect energy minimizer can be improved.

References

[BG25] Ahmed Bou-Rabee and Ewain Gwynne. *Random walk on sphere packings and Delaunay triangulations in arbitrary dimension.* arXiv:2405.11673v2, 2025.